Control of Mobile Robots - Laboratory session 3
An odeint robot simulator
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Develop a ROS 2 aperiodic node implementing a robot simulator.
The robot is described by the unicycle kinematic model

$$\begin{aligned}\dot{x} &= v\cos\theta \\ \dot{y} &= v\sin\theta \\ \dot{\theta} &= \omega\end{aligned}$$

where x, y, and θ denote the robot position and orientation, while v and ω denote its linear and angular velocities, respectively.
The simulator receives the linear and angular velocity commands on the `/cmd_vel` topic and publishes:

- the robot odometry (position, orientation, and velocities) on `/odom`;
- the simulation time on `/clock`.

The simulator must support different publishing frequencies for the odometry and clock messages. These frequencies must be configurable through two ROS 2 parameters. The initial robot pose must also be configurable through appropriate parameters.

Develop a second ROS 2 node to generate the velocity commands required to test the simulator. This node must be periodic and publish the linear and angular velocity commands on `/cmd_vel`.
The commands must drive the robot along a figure-eight trajectory described by

$$\begin{aligned}x(t) &= a\sin\left(\frac{2\pi}{T}t\right) \\ y(t) &= a\sin\left(\frac{2\pi}{T}t\right)\cos\left(\frac{2\pi}{T}t\right)\end{aligned}$$

where a determines the amplitude of the trajectory and T is the time required to complete one lap.
Using the differential-flatness property of the unicycle model, compute the feedforward velocity commands as

$$\begin{aligned}v(t) &= \sqrt{\dot{x}(t)^2+\dot{y}(t)^2} \\ \omega(t) &= \frac{\dot{x}(t)\ddot{y}(t)-\dot{y}(t)\ddot{x}(t)}{\dot{x}(t)^2+\dot{y}(t)^2}\end{aligned}$$

The trajectory parameters a and T must be implemented as ROS 2 parameters of the command-generator node.

Finally, provide:

- a ROS 2 launch file that starts one instance of the simulator node and one instance of the command-generator node;
- a Python script for reading and analysing the data generated by the simulator.

Test the complete system by recording the relevant topics in a ROS 2 bag. Verify the correct behaviour of the simulator by plotting:

- the robot trajectory in the x-y plane;
- the linear velocity command $v(t)$;
- the angular velocity command $\omega(t)$.